

PYTHON

```
# Accepting user inputs
name = input("Enter your name: ")
# Converting age to an integer to demonstrate different types
age = int(input("Enter your age: "))

# Printing a formatted message using an f-string
print(f"\nHello {name}! You are {age} years old.")

# Displaying the datatype of each input
print(f"The datatype of 'name' is: {type(name)}")
print(f"The datatype of 'age' is: {type(age)}")
```

```
Enter your name: krishan
```

```
Enter your age: 18
```

```
Hello krishan! You are 18 years old.
```

```
The datatype of 'name' is: <class 'str'>
```

```
The datatype of 'age' is: <class 'int'>
```

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=== Code Execution Successful ===
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```
# Taking input from the user
# We use float() to allow decimal numbers
num = float(input("Enter a number: "))
```

```
# Checking conditions
if num > 0:
    print("The number is positive.")
elif num < 0:
    print("The number is negative.")
else:
```

```
    print("The number is zero.")
```

```
Enter a number: 17
The number is positive.

=== Code Execution Successful ===
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```
# Program to calculate the sum of first n natural numbers
n = int(input("Enter a positive integer: "))
```

```
sum_total = 0
```

```
# Use range(1, n + 1) to include the number n
for i in range(1, n + 1):
    sum_total += i
```

```
print(f"The sum of the first {n} natural numbers is: {sum_total}")
```

```
Enter a positive integer: 2
```

```
The sum of the first 2 natural numbers is: 3
```

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=== Code Execution Successful ===
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```

def demonstrate_operators():
    # Initializing variables
    a, b = 15, 4
    x, y = True, False

    # 1. Arithmetic Operators: Used for mathematical calculations
    print("--- Arithmetic Operators ---")
    print(f"Addition: {a} + {b} = {a + b}")
    print(f"Subtraction: {a} - {b} = {a - b}")
    print(f"Multiplication: {a} * {b} = {a * b}")
    print(f"Division (Float): {a} / {b} = {a / b}")
    print(f"Floor Division: {a} // {b} = {a // b}") # Returns whole number part
    print(f"Modulus (Remainder): {a} % {b} = {a % b}")
    print(f"Exponentiation: {a} ** {b} = {a ** b}") # a to the power of b

    # 2. Comparison Operators: Evaluate to True or False
    print("\n--- Comparison Operators ---")
    print(f"Is Equal: {a} == {b} is {a == b}")
    print(f"Not Equal: {a} != {b} is {a != b}")
    print(f"Greater Than: {a} > {b} is {a > b}")
    print(f"Less Than: {a} < {b} is {a < b}")
    print(f"Greater Than or Equal: {a} >= 15 is {a >= 15}")

    # 3. Logical Operators: Combine multiple conditions
    print("\n--- Logical Operators ---")
    print(f"AND (both True): {x} and {y} is {x and y}")
    print(f"OR (at least one True): {x} or {y} is {x or y}")
    print(f"NOT (inverse): not {x} is {not x}")

if __name__ == "__main__":

```

```
--- Arithmetic Operators ---
Addition: 15 + 4 = 19
Subtraction: 15 - 4 = 11
Multiplication: 15 * 4 = 60
Division (Float): 15 / 4 = 3.75
Floor Division: 15 // 4 = 3
Modulus (Remainder): 15 % 4 = 3
Exponentiation: 15 ** 4 = 50625

--- Comparison Operators ---
Is Equal: 15 == 4 is False
Not Equal: 15 != 4 is True
Greater Than: 15 > 4 is True
Less Than: 15 < 4 is False
Greater Than or Equal: 15 >= 15 is True

--- Logical Operators ---
AND (both True): True and False is False
OR (at least one True): True or False is True
NOT (inverse): not True is False

=== Code Execution Successful ===
```

demonstrate_operators()

```

def analyze_numbers():
    try:
        # Prompt the user for input and split by spaces
        user_input = input("Enter numbers separated by spaces: ")

        # Convert the string inputs into a list of floats
        numbers = [float(x) for x in user_input.split()]

        # Check if the list is empty to avoid errors
        if not numbers:
            print("No numbers were entered.")
            return

        # Perform calculations using built-in functions
        maximum = max(numbers) # Finds the highest value
        minimum = min(numbers) # Finds the lowest value
        total_sum = sum(numbers) # Calculates the total sum

        # Sort the list (creates a new sorted version)
        sorted_list = sorted(numbers)

        # Output the results
        print(f"\n--- Results ---")
        print(f"Maximum: {maximum}")
        print(f"Minimum: {minimum}")
        print(f"Sum: {total_sum}")
        print(f"Sorted List: {sorted_list}")

    except ValueError:
        print("Error: Please enter only valid numbers.")

```

```

if __name__ == "__main__":

```

```

    Enter numbers separated by spaces: 1 2 3 4

    --- Results ---
    Maximum: 4.0
    Minimum: 1.0
    Sum: 10.0
    Sorted List: [1.0, 2.0, 3.0, 4.0]

    === Code Execution Successful ===

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    analyze_numbers()

```

```

def process_string(text):
    # 1. Count Vowels
    vowels = "aeiouAEIOU"
    vowel_count = sum(1 for char in text if char in vowels)

    # 2. Reverse the String
    reversed_text = text[::-1]

    # 3. Check for Palindrome
    # We clean the text by removing spaces and making it lowercase for a true check
    cleaned_text = "".join(char.lower() for char in text if char.isalnum())
    is_palindrome = cleaned_text == cleaned_text[::-1]

    # Display the results
    print("\n--- Results ---")
    print(f"Original String: {text}")
    print(f"Number of Vowels: {vowel_count}")
    print(f"Reversed String: {reversed_text}")
    print(f"Is Palindrome?: {'Yes' if is_palindrome else 'No'}")

# Accept input from the user
user_input = input("Enter a string to analyze: ")

```

```
Enter a string to analyze: 455
```

```
--- Results ---
```

```
Original String: 455
```

```
Number of Vowels: 0
```

```
Reversed String: 554
```

```
Is Palindrome?: No
```

```
=== Code Execution Successful ===
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```
process_string(user_input)
```

```

def get_factorial(n):
    """Calculates the factorial of a non-negative integer using iteration."""
    if n < 0:
        return "Factorial is not defined for negative numbers."
    result = 1
    for i in range(1, n + 1):
        result *= i
    return result

def get_fibonacci(n_terms):
    """Generates a Fibonacci sequence up to the specified number of terms."""
    if n_terms <= 0:
        return []
    elif n_terms == 1:
        return [0]

    sequence = [0, 1]
    while len(sequence) < n_terms:
        # The next number is the sum of the previous two
        next_val = sequence[-1] + sequence[-2]
        sequence.append(next_val)
    return sequence

# --- Example Usage ---
num = 5
print(f"Factorial of {num}: {get_factorial(num)}")

terms = 10
print(f"Fibonacci sequence ({terms} terms): {get_fibonacci(terms)}")

```

```

Factorial of 5: 120
Fibonacci sequence (10 terms): [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]

=== Code Execution Successful ===

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```

# Initialize an empty dictionary to store student records
student_data = {}

# Get the number of students to add
num_students = int(input("Enter the number of students: "))

# Loop to accept student name and marks
for _ in range(num_students):
    name = input("Enter student name: ")
    marks = float(input(f"Enter marks for {name}: "))
    student_data[name] = marks

# Display the dictionary
print("\n--- Student Records ---")
for student, marks in student_data.items():
    print(f"{student}: {marks}")

# Check if the dictionary is not empty to avoid errors
if student_data:
    # Find the highest and lowest scorers based on the values
    highest_scorer = max(student_data, key=student_data.get)
    lowest_scorer = min(student_data, key=student_data.get)

    print("\n--- Results ---")
    print(f"Highest Scorer: {highest_scorer} ({student_data[highest_scorer]} marks)")
    print(f"Lowest Scorer: {lowest_scorer} ({student_data[lowest_scorer]} marks)")
else:

```

```

Enter the number of students: 1
Enter student name: krishan
Enter marks for krishan: 100

--- Student Records ---
krishan: 100.0

--- Results ---
Highest Scorer: krishan (100.0 marks)
Lowest Scorer: krishan (100.0 marks)

=== Code Execution Successful ===

```

```

print("No student data found.")

```